Unit 2 Test Study Guide

Expressions, Equations, and Inequalities | New Format

Name: ______________________________ **Period:** __________ **Due:** ____________

How to use this guide: This guide follows the Unit 2 test structure while giving each skill a Version A and Version B. Try each problem before using the self-check bank at the end.

Multiple-Choice Practice

1A. Order of Operations | Version A

What is the value of $8 \div 4 - 3 \cdot 2 + 12$? A. −16 B. 2 C. 8 D. 14	Work / thinking __________________________ __________________________ __________________________

1B. Order of Operations | Version B

What is the value of $18 - 6 \div 3 \cdot 4 + 5$? A. −1 B. 15 C. 21 D. 29	Work / thinking __________________________ __________________________ __________________________

2A. Translating Expressions | Version A

Which expression represents “five less than twice a number”? A. $n - 5 \cdot 2$ B. $5 - 2n$ C. $2(n - 5)$ D. $2n - 5$	Work / thinking __________________________ __________________________ __________________________

2B. Translating Expressions | Version B

Which expression represents “seven more than three times a number”? A. $3n + 7$ B. $7n + 3$ C. $3(n + 7)$ D. $7 - 3n$	Work / thinking __________________________ __________________________ __________________________

3A. Evaluating Expressions | Version A

Evaluate $(a^2 - 2b)/c$ when $a = -3$, $b = 4$, and $c = -1$. A. −1 B. −17 C. 17 D. 1	Work / thinking ________________________ ________________________ ________________________

3B. Evaluating Expressions | Version B

Evaluate $(m^2 + 3n)/p$ when $m = -2$, $n = 5$, and $p = -1$. A. 19 B. −11 C. −19 D. 11	Work / thinking ________________________ ________________________ ________________________

4A. Solving Equations | Version A

Solve $4(2x - 1) = -3(x + 5)$. A. $x = -19/5$ B. $x = -1$ C. $x = -11$ D. $x = 1$	Work / thinking ________________________ ________________________ ________________________

4B. Solving Equations | Version B

Solve $3(2x + 5) = -2(x - 1)$. A. $x = -13/8$ B. $x = 13/8$ C. $x = -17/4$ D. $x = -1$	Work / thinking ________________________ ________________________ ________________________

5A. Number of Solutions | Version A

How many solutions does $3(x + 2) + 2 = 2(x + 2) + 4x + 2$ have? A. One solution B. Two solutions C. Infinitely many solutions D. No solution	Work / thinking ________________________ ________________________ ________________________

5B. Number of Solutions | Version B

How many solutions does $2(x - 3) + 4 = 2x - 2$ have? A. One solution B. Two solutions C. Infinitely many solutions D. No solution	Work / thinking ________________________ ________________________ ________________________

6A. Equations with Fractions | Version A

Solve $(2/3)x - 1/3 = 1/2$. A. $x = 5/4$ B. $x = 1/4$ C. $x = 3/5$ D. $x = 5/9$	Work / thinking ________________________ ________________________ ________________________

6B. Equations with Fractions | Version B

Solve $(3/4)x + 1/2 = 2$. A. $x = 3/2$ B. $x = 2$ C. $x = 8/3$ D. $x = 10/3$	Work / thinking ________________________ ________________________ ________________________

7A. Literal Equations | Version A

Solve $4T + 4M = C$ for M. A. $M = C - T$ B. $M = (C - T)/4$ C. $M = C/4 - T$ D. $M = C - 4T$	Work / thinking ______________________________ ______________________________ ______________________________

7B. Literal Equations | Version B

Solve $3x - 6y = 12$ for y. A. $y = 2 - x/2$ B. $y = x/2 - 2$ C. $y = 2x - 6$ D. $y = 12 - 3x$	Work / thinking ______________________________ ______________________________ ______________________________

8A. Testing Inequality Values | Version A

Select all values that make $-5x < 50$ true. A. −10 B. 10 C. −50 D. −5 E. 50	Work / thinking ______________________________ ______________________________ ______________________________

8B. Testing Inequality Values | Version B

Select all values that make $4x - 3 \geq 9$ true. A. -2 B. 0 C. 3 D. 4 E. 6	Work / thinking ____________________________ ____________________________ ____________________________

9A. Inequality Representations | Version A

Select the correct representation of $2x - 5 \leq 4x - 1$. A. $x \leq 2$ B. $x \leq -2$ C. $x \geq 2$ D. $x \geq -2$	Work / thinking ____________________________ ____________________________ ____________________________

9B. Inequality Representations | Version B

Select the correct representation of $-3x + 4 < 10$. A. $x < -2$ B. $x > -2$ C. $x < 2$ D. $x > 2$	Work / thinking ____________________________ ____________________________ ____________________________

Free-Response Practice

Show clear work. For inequalities, include inequality notation and interval notation when requested.

10-A. Representing Inequalities | Version A

Complete each missing representation. a. $x > 4$ → interval notation: ________ b. $x \le 5$ → interval notation: ________ c. $(-\infty, 0) \cup [2, \infty)$ → inequality notation: ________ **Answer:** ____________________	Work / thinking ______________________________ ______________________________ ______________________________ ______________________________ ______________________________

10-B. Representing Inequalities | Version B

Complete each missing representation. a. $x \ge -3$ → interval notation: ________ b. $x < 2$ → interval notation: ________ c. $(1, 5]$ → inequality notation: ________ **Answer:** ____________________	Work / thinking ______________________________ ______________________________ ______________________________ ______________________________ ______________________________

11-A. Order of Operations | Version A

Evaluate $\lvert 2 - 4\rvert + 3 \cdot 2^2 - \lvert 1 - 4\rvert$. **Answer:** ____________________	Work / thinking ______________________________ ______________________________ ______________________________ ______________________________ ______________________________

11-B. Order of Operations | Version B

Evaluate $\lvert 5 - 9\rvert + 2 \cdot 3^2 - \lvert 7 - 10\rvert$. **Answer:** ____________________	Work / thinking ______________________________ ______________________________ ______________________________ ______________________________ ______________________________

12a-A. Writing Expressions | Version A

Write an algebraic expression: Twice the sum of a number and eleven. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

12a-B. Writing Expressions | Version B

Write an algebraic expression: Three times the difference of a number and four. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

12b-A. Writing Expressions | Version A

Write an algebraic expression: Five less than the product of a number and six. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

12b-B. Writing Expressions | Version B

Write an algebraic expression: Nine more than the product of a number and two. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

13a-A. Evaluating Expressions | Version A

Evaluate $x^2 - 4x - 7$ for $x = -2$. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

13a-B. Evaluating Expressions | Version B

Evaluate $x^2 + 3x - 10$ for $x = -4$. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

13b-A. Evaluating Expressions | Version A

Evaluate $-2x + 5y$ for $x = -2$ and $y = 3$. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

13b-B. Evaluating Expressions | Version B

Evaluate $4a - 3b$ for $a = -1$ and $b = 5$. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

14a-A. Equations from Context | Version A

Parking costs \$10.25 and each ticket costs \$14.25. A group has \$152.75 total. Write and solve an equation for x, the number of people who can go. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

14a-B. Equations from Context | Version B

Shoe rental costs \$7.50 and each bowling game costs \$5.25. The total is \$49.50. Write and solve an equation for x, the number of games. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

14b-A. Consecutive Integer Equations | Version A

The sum of three consecutive integers is 93. Write and solve an equation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

14b-B. Consecutive Integer Equations | Version B

The sum of three consecutive integers is 126. Write and solve an equation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

15a-A. Solving Equations | Version A

Solve $8 - 3x + 2 = -3(x + 3) + 21$. **Answer:** ____________________	Work / thinking ______________________________ ______________________________ ______________________________ ______________________________ ______________________________ ______________________________

15a-B. Solving Equations | Version B

Solve $5 - 2x + 7 = -2(x - 4) + 4$. **Answer:** ____________________	Work / thinking ______________________________ ______________________________ ______________________________ ______________________________ ______________________________ ______________________________

15b-A. Solving Equations | Version A

Solve $2(x - 1) + 4x = -x - 2 + 7x$. **Answer:** ____________________	Work / thinking ______________________________ ______________________________ ______________________________ ______________________________ ______________________________

15b-B. Solving Equations | Version B

Solve $4(x + 3) - 2x = 2x + 7$. **Answer:** ____________________	Work / thinking ______________________________ ______________________________ ______________________________ ______________________________ ______________________________ ______________________________

15c-A. Solving Equations | Version A

Solve $6x + 5 = 1/2(8x + 4)$. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

15c-B. Solving Equations | Version B

Solve $3x - 4 = 1/3(6x + 9)$. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

16a-A. Literal Equations | Version A

Solve $2t + w = p$ for t. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

16a-B. Literal Equations | Version B

Solve $3r + s = m$ for r. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

16b-A. Literal Equations | Version A

Solve $3x - 6y = 12$ for y. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

16b-B. Literal Equations | Version B

Solve $5a + 2b = 20$ for b. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

17a-A. Solving Inequalities | Version A

Solve $2x - 4 \leq 4$. Write inequality and interval notation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

17a-B. Solving Inequalities | Version B

Solve $3x + 5 > 14$. Write inequality and interval notation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

17b-A. Solving Inequalities | Version A

Solve $3x - 6 < 8x - 1$. Write inequality and interval notation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

17b-B. Solving Inequalities | Version B

Solve $-4x + 7 \leq 15$. Write inequality and interval notation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

17c-A. Compound Inequalities | Version A

Solve $-16 < 2x - 4 \leq 4$. Write inequality and interval notation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

17c-B. Compound Inequalities | Version B

Solve $2 \leq 3x - 7 < 8$. Write inequality and interval notation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

17d-A. Compound Inequalities | Version A

Solve $2x + 2 < 6$ or $x - 6 \geq -1$. Write inequality and interval notation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

17d-B. Compound Inequalities | Version B

Solve $x + 3 \leq -1$ or $x + 1 \geq 3$. Write inequality and interval notation. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

Student Self-Check Answer Bank

Important: This is not a worked solution key. It is a self-check bank. If your answer is not here, go back and check the rule, the sign, and the equation or inequality step.

Multiple-Choice Practice

Solve first, then check. Cross off each answer only after your work supports it.

D. $x \geq -2$	B. 15
A. $x = -13/8$	C, D, E
C. −19	B. $x = -1$
D. $2n - 5$	A. One solution
C. Infinitely many solutions	B. $x = 2$
B. $y = x/2 - 2$	C. 8
A. $3n + 7$	A. −1
A. $x = 5/4$	B, D, E
B. $x > -2$	C. $M = C/4 - T$

Free-Response Practice

Solve first, then check. Cross off each answer only after your work supports it.

$3(n - 4)$	$x > 3$; $(3, \infty)$
$t = (p - w)/2$	$-6 < x \leq 4$; $(-6, 4]$
No solution	$2(n + 11)$
19	10 people
−19	$x = 7$
a. $[-3, \infty)$; b. $(-\infty, 2)$; c. $1 < x \leq 5$	5
$x \geq -2$; $[-2, \infty)$	$x \leq -4$ or $x \geq 2$; $(-\infty, -4] \cup [2, \infty)$
$2n + 9$	41, 42, 43
Infinitely many solutions	a. $(4, \infty)$; b. $(-\infty, 5]$; c. $x < 0$ or $x \geq 2$
$6n - 5$	8 games
$y = x/2 - 2$	$x = -3/2$
$x < 2$ or $x \geq 5$; $(-\infty, 2) \cup [5, \infty)$	30, 31, 32
−6	11
$x > -1$; $(-1, \infty)$	$3 \leq x < 5$; $[3, 5)$
$b = 10 - (5/2)a$	19
$x \leq 4$; $(-\infty, 4]$	Infinitely many solutions
−19	$r = (m - s)/3$

Final Check Before the Test

- ☐ I can apply order of operations accurately.
- ☐ I can translate phrases into algebraic expressions.
- ☐ I can solve equations and identify no-solution or infinitely-many-solution cases.
- ☐ I can solve literal equations for a requested variable.
- ☐ I can solve and represent inequalities using interval notation.